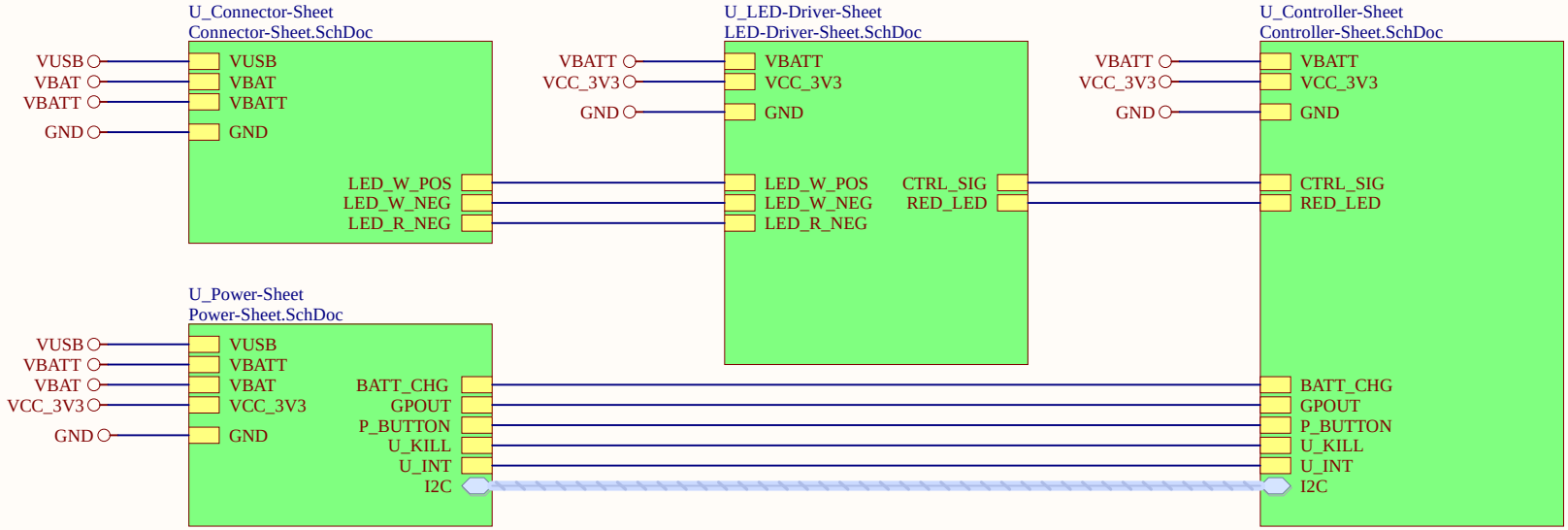
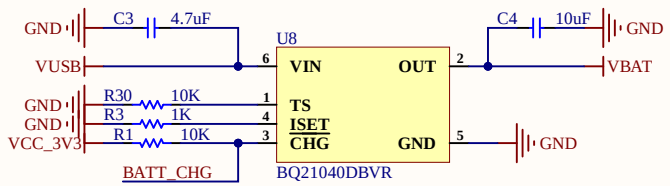


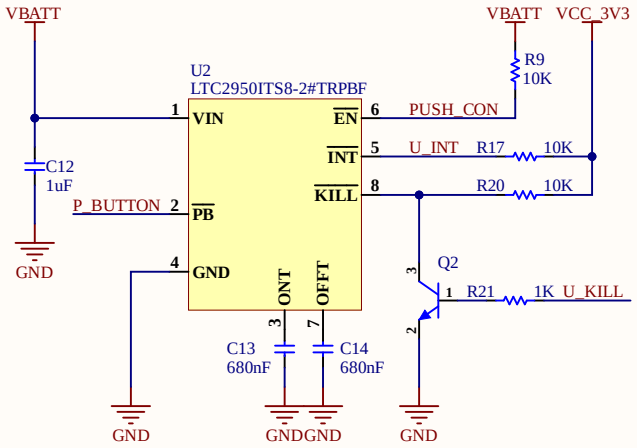
Title		
Size: A	Number	Revision
Date: 11/18/2024		Sheet of
File: C:\Users\..\Top-Sheet.SchDoc		Drawn By:

BATTERY CHARGER

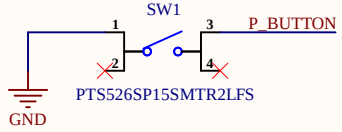
540mA Charging Current



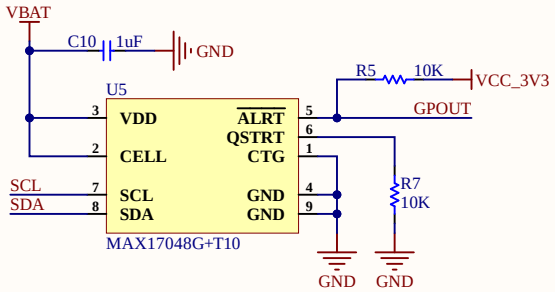
ON/OFF CONTROLLER



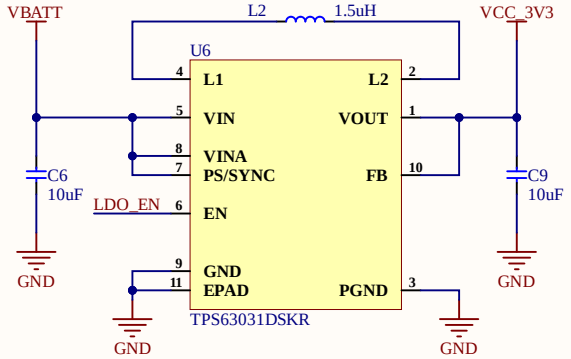
MODE/POWER BUTTON



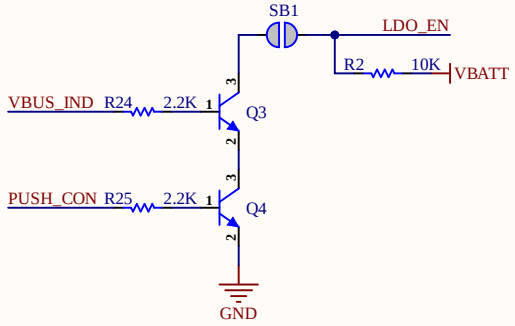
BATTERY FUEL GAUGE



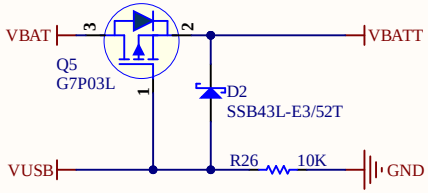
3V3 LDO

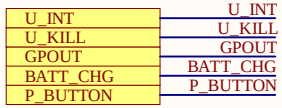


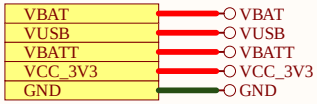
BJT NAND GATE



Oring Circuit









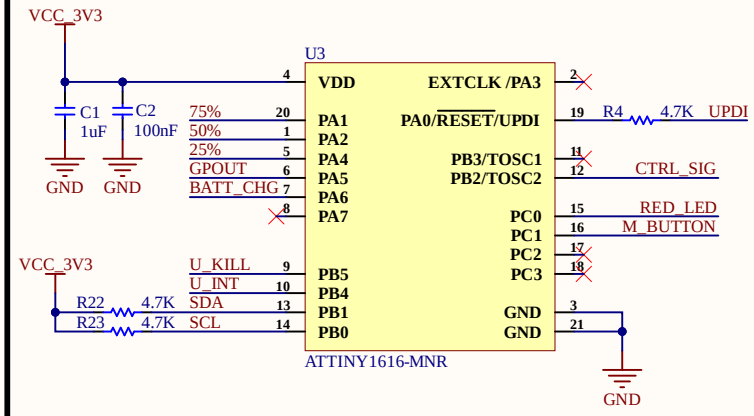
eSpark Consulting Group

Title: Power-Sheet.SchDoc

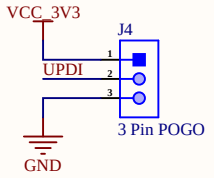
File: Control-Board_v2.4.PrjPcb

Date: 18/11/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 5

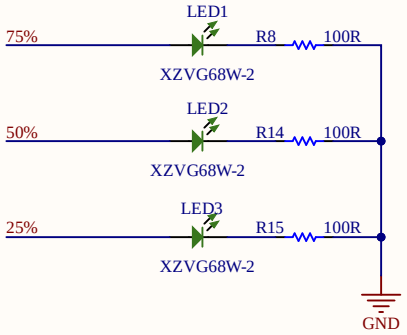
ATTINY1616



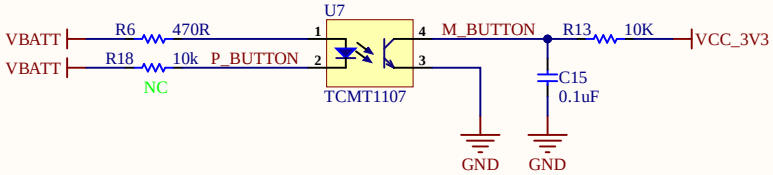
PROGRAMMING

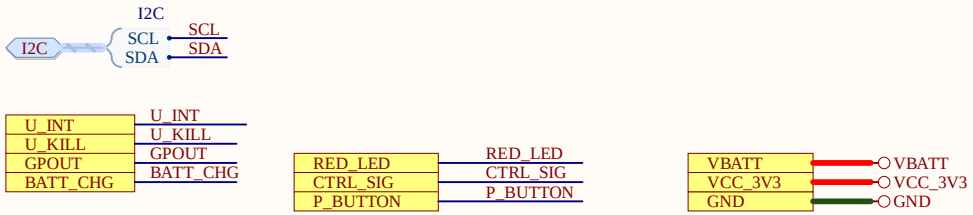


BATTERY LEVEL



MODE SELECTION

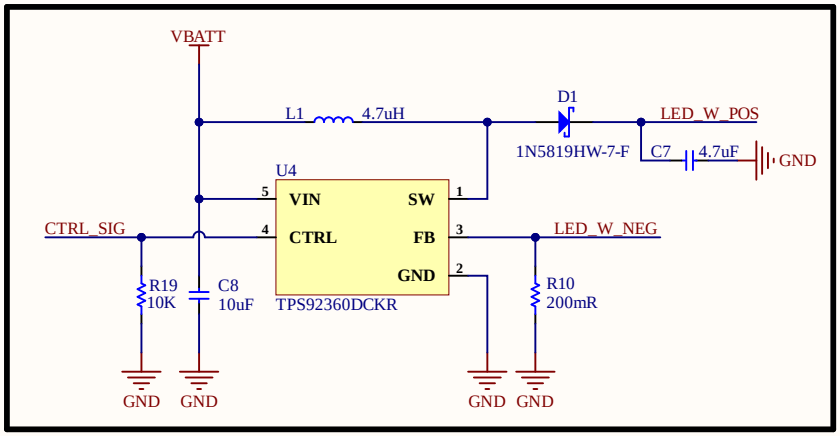




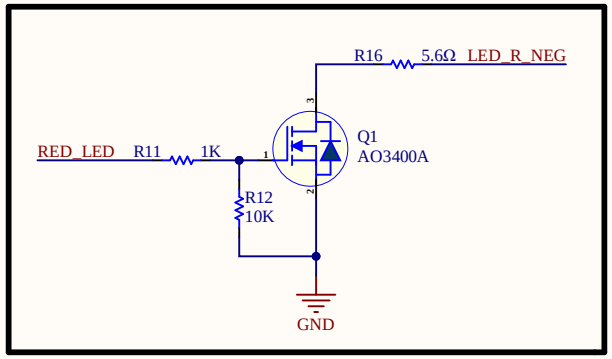
eSpark Consulting Group

Title: Controller-Sheet.SchDoc	
File: Control-Board_v2.4.PrjPcb	
Date: 13/11/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 5

LED DRIVER



RED LED CONTROL



LED_W_POS — LED_W_POS
LED_W_NEG — LED_W_NEG
LED_R_NEG — LED_R_NEG

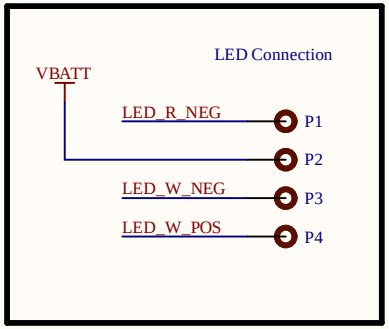
RED_LED — RED_LED
CTRL_SIG — CTRL_SIG

VBATT — VBATT
VCC_3V3 — VCC_3V3
GND — GND

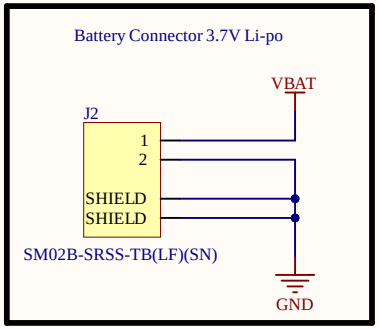
eSpark Consulting Group

Title: LED-Driver-Sheet.SchDoc	
File: Control-Board_v2.4.PrjPcb	
Date: 13/11/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 5

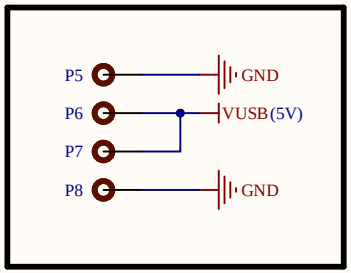
LED INTERFACE CONNECTOR

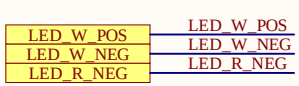


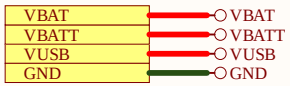
BATTERY INPUT



USB INTERFACE CONNECTOR

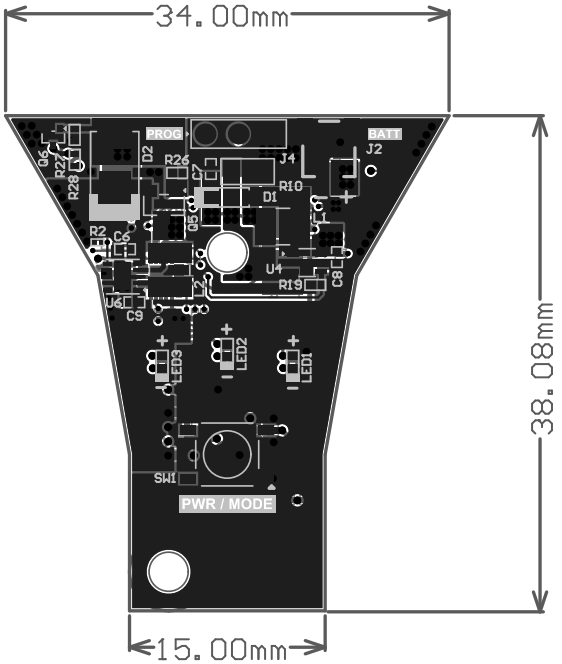






eSpark Consulting Group

Title: Connector-Sheet.SchDoc	
File: Control-Board_v2.4.PrjPcb	
Date: 13/11/2024	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 5



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.010mm	3.5	
1	Top Layer		0.035mm		
	Dielectric 2	PP-006	0.200mm	4.1	
2	GND	CF-004	0.035mm		
	Dielectric 1	FR-4	1.030mm	4.8	
3	POWER	CF-004	0.035mm		
	Dielectric 3	PP-006	0.200mm	4.1	
4	Bottom Layer		0.035mm		
	Bottom Solder	Solder Resist	0.010mm	3.5	
	Bottom Overlay				

Total board thickness: 1.590mm